Joint Dwell Time Allocation and Route Planning for Targets Tracking in Radar Network via Deep Reinforcement Learning

Zhixian Wu, Yang Su*
School of Electronic and Information Engineering
Nanjing University of Information Science and Technology
Nanjing, China
ysu@nuist.edu.cn

***Abstract*—This paper proposes a cognitive cooperative tracking architecture integrating a bidirectional long short-term memory (Bi-LSTM) network with Q-learning to address resource and geometry challenges in multi-radar systems. The Bi-LSTM module leverages temporal attention to anticipate maneuver trends by monitoring prediction-filter deviations, while the Q-learning agent jointly optimizes beam dwell time and radar platform motion within a Markov Decision Process(MDP) framework. This strategy improves Signal-to-Noise Ratio (SNR) and observation geometry by autonomously guiding platforms toward high-uncertainty targets. Simulation results demonstrate that this joint control significantly reduces tracking error and target loss rate in complex scenarios compared to static uniform allocation baselines.**

***Index Terms*—cognitive radar, Q-Learning, Deep Learning, IMM.**

I. Introduction

Modern radar systems face severe challenges in maintaining track continuity due to increasing electromagnetic complexity and target agility [1]. Cognitive Radar (CR), inspired by biological perception-action cycles [1][2], enables systems to actively learn and optimize transmission strategies. Reinforcement Learning (RL) has emerged as a dominant paradigm for these sequential decision-making problems , allowing radars to derive optimal policies from experience. Recent works have extensively explored RL for various tasks, such as Lu et al. who proposed a Constrained DQN to optimize time allocation [3], extended it to scanning [4], and utilized DDPG for continuous action spaces [5]. RL has also proved effective in waveform parameter selection [6][7], multi-objective trade-offs [8], anti-jamming [9], and spectral interference mitigation [10].

However, most existing resource scheduling strategies rely heavily on state estimation accuracy. Traditional Interacting Multiple Model (IMM) filters often suffer from "model lag" when targets execute abrupt maneuvers. While data-driven methods like RNNs [11], Track-LITE [12], and Bi-LSTM [13] have been utilized to capture non-linear motion and refine estimates, maneuver perception and resource allocation are typically treated as decoupled tasks. Furthermore, existing frameworks primarily focus on electronic resource allocation assuming static platforms, overlooking mobility as a controllable resource. Recent studies suggest that optimizing the physical positions of mobile agents can fundamentally improve observation geometry [14][15] and SNR.

We propose a cognitive cooperative tracking architecture integrating Bi-LSTM active perception with joint resource-motion decision-making. The Bi-LSTM module anticipates maneuvers to compensate for IMM lag, while a Q-learning agent within an MDP framework optimizes beam dwell time and platform velocity vectors. Numerical results confirm this closed-loop mechanism significantly reduces tracking error and target loss rate compared to static uniform baselines.

II. System Model

We consider a 2D dynamic tracking scenario involving distributed N cognitive radars and M targets with distinct maneuvering characteristics

A. Target Dynamics

The j-th target state vector at time step k is defined as $\mathbf{X}_k^{(j)} = [x_k^{(j)}, v_{x,k}^{(j)}, y_k^{(j)}, v_{y,k}^{(j)}]^T$. The position vector of the j-th target is $\mathbf{P}_{t,k}^{(j)} = [x_k^{(j)}, y_k^{(j)}]^T$.

B. Radar Observation Model

The position of the i-th radar at time k is $\mathbf{P}_{r,k}^{(i)} = [x_{r,k}^{(i)}, y_{r,k}^{(i)}]^T$. The measurement vector $\mathbf{Z}_{r,k}^{(i)}$ consists of range r and azimuth θ:

$$\mathbf{Z}_{r,k}^{(i)} = h\left(\mathbf{P}_{t,k}^{(j)}, \mathbf{P}_{r,k}^{(i)}\right) + \mathbf{v}_k^{(i)} \tag{1}$$

where $h(\cdot)$ is the nonlinear observation function, and $\mathbf{v}_k^{(i)}$ is the zero-mean Gaussian measurement noise with covariance $\mathbf{R}_k^{(i)} = \text{diag}(\sigma_{r,k}^2, \sigma_{\theta,k}^2)$, with angular noise standard deviation $\sigma_{\theta,k} = 0.1°$.

Based on the radar equation, the range measurement accuracy is physically constrained by the beam dwell time $t_k^{(i,j)}$ and the radar-to-target distance U_k. The noise standard deviation $\sigma_{r,k}$ is modeled as:

$$\sigma_{r,k} = \sqrt{\sigma_{sys}^2 + \left(\frac{\sigma_{base}}{\sqrt{t_k^{(i,j)}}} \cdot \left(\frac{U_k}{R_{ref}}\right)^2\right)^2} \tag{2}$$

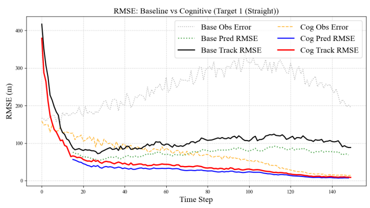

Fig. 1. RMSE analysis for Target 1 (Steady).

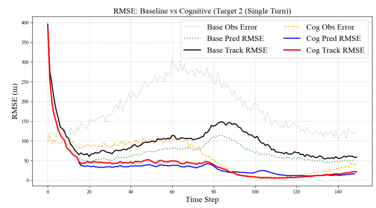

Fig. 2. RMSE analysis for Target 2 (Single Maneuver).

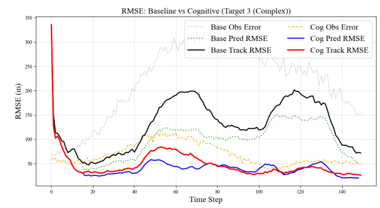

Fig. 3. RMSE analysis for Target 3 (Complex Maneuver).

where $\sigma_{sys} = 10$ m is the system noise floor, $\sigma_{base} = 450$ m is the baseline noise coefficient, and $R_{ref} = 10000$ m is the reference range.

At each time step k, the distributed measurements $\{\mathbf{Z}_{r,k}^{(1)}, \dots, \mathbf{Z}_{r,k}^{(N)}\}$ from all N radars are transmitted to the fusion center. We employ the Covariance Intersection (CI) algorithm to integrate these local observations into a single Global Fused Measurement vector $\mathbf{Z}_{t,k}^{(j)}$ for each target j.

C. Radar Platform Kinematics

The radar platforms are modeled as controllable point masses. The discrete-time kinematic update is given by:

$$\mathbf{P}_{r,k+1}^{(i)} = \mathbf{P}_{r,k}^{(i)} + \mathbf{v}_{r,k}^{(i)} \cdot \Delta t \tag{3}$$

where $\mathbf{v}_{r,k}^{(i)}$ is determined by the cognitive decision module detailed in Section III-B, and $\Delta t = 1$ s.

D. IMM Framework

The tracker assumes the target state vector is defined as $\widehat{\mathbf{X}}_k^{(j)} = [\hat{x}_k^{(j)}, \hat{v}_{x,k}^{(j)}, \hat{y}_k^{(j)}, \hat{v}_{y,k}^{(j)}]^T$, and the estimated position vector of the j-th target is $\widehat{\mathbf{P}}_{t,k}^{(j)} = [\hat{x}_k^{(j)}, \hat{y}_k^{(j)}]^T$.

The state transition is governed by:

$$\widehat{\mathbf{X}}_{k+1}^{(j)} = \mathbf{F}_k^{(m)} \widehat{\mathbf{X}}_k^{(j)} + \mathbf{w}_k^{(m)} \tag{4}$$

where $\mathbf{F}_k^{(m)}$ is the transition matrix and $\mathbf{w}_k^{(m)}$ is the process noise.

III. Proposed Q-Learning Based Dwell Time Allocation and Route Planning Algorithm

The proposed cognitive tracking architecture integrates two modules: a Bi-LSTM active perception module for maneuver anticipation, and a Q-Learning agent for joint dwell time allocation and radar platform motion control.

A. Bi-LSTM

The IMM filter outputs a 4-dimensional state vector $\widehat{\mathbf{X}}_k^{(j)}$ containing only position and velocity. However, explicit acceleration information is critical for identifying maneuver onset. Therefore, we construct an augmented 6-dimensional feature vector $\mathbf{f}_k^{(j)}$ at time step k. This vector includes the estimated position, velocity, and the computed acceleration derived from second-order discrete differences.Although acceleration in the input is kinematically redundant with velocity, its explicit inclusion serves as a direct physical cue for the Temporal Attention mechanism:

$$\mathbf{f}_k^{(j)} = \left[\hat{x}_k^{(j)}, \hat{y}_k^{(j)}, \hat{v}_{x,k}^{(j)}, \hat{v}_{y,k}^{(j)}, \hat{a}_{x,k}^{(j)}, \hat{a}_{y,k}^{(j)}\right]^T \tag{5}$$

where the acceleration components are calculated as $\hat{a}_{x,k}^{(j)} = (\hat{v}_{x,k}^{(j)} - \hat{v}_{x,k-1}^{(j)})/\Delta t$ and $\hat{a}_{y,k}^{(j)} = (\hat{v}_{y,k}^{(j)} - \hat{v}_{y,k-1}^{(j)})/\Delta t$.

The input to the network is a sliding window of these features with length L, denoted as sequence $\mathcal{F}_k^{(j)} = \{\mathbf{f}_{k-L+1}^{(j)}, \dots, \mathbf{f}_k^{(j)}\} \in \mathbb{R}^{L \times 6}$.

We employ a 2-layer Bi-LSTM network with a hidden dimension of $d_h = 256$ to process the sequence $\mathcal{F}_k^{(j)}$ of length $L = 16$,which is outlined in Algorithm 1. The input dimension $d_{in} = 6$ is mapped to a 2-dimensional predicted position $d_{in} = 2$.To prevent overfitting, a dropout rate of $p_{drop} = 0.2$ is applied during training.

Let $\mathbf{h}_n = [\overrightarrow{\mathbf{h}}_n; \overleftarrow{\mathbf{h}}_n] \in \mathbb{R}^{1 \times 2d_h}$ denote the concatenated hidden state for step $n \in \{1, \dots, L\}$. To identify maneuvers, a temporal attention mechanism computes scalar scores $e_n = \mathbf{W}_{att}\mathbf{h}_n + b_{att}$, normalized weights $\alpha_n = e^{e_n} / \sum_{j=1}^{L} e^{e_j}$, and a context vector $\mathbf{c}_k = \sum_{n=1}^{L} \alpha_n \mathbf{h}_n$.

Subsequently, the predicted future position is generated as $\hat{\mathbf{P}}_{lstm,k+1}^{(j)} = \mathbf{W}_{fc}\mathbf{c}_k + \mathbf{b}_{fc}$. Finally, we define $\delta_k^{(j)} = \|\hat{\mathbf{P}}_{lstm,k+1}^{(j)} - \hat{\mathbf{P}}_{t,k+1}^{(j)}\|_2$ to quantify the discrepancy between the deep learning prediction and the IMM filter estimation.

Algorithm 1 Bi-LSTM Prediction (for each j target)

Require: $\mathcal{F}_k^{(j)} = \{\mathbf{f}_{k-L+1}^{(j)}, \dots, \mathbf{f}_k^{(j)}\}$
Ensure: $\widehat{\mathbf{P}}_{lstm,k+1}^{(j)}, \delta_k^{(j)}$

Step 1: Bi-LSTM Feature Extraction
Normalize input: $\bar{\mathcal{F}}_k \leftarrow \text{Norm}(\mathcal{F}_k^{(j)})$
for time step $n = 1$ to L **do**
  Forward $\overrightarrow{\mathbf{h}}_n \leftarrow \text{LSTM}_{fwd}[\bar{\mathbf{f}}_n, \overrightarrow{\mathbf{h}}_{n-1}]$
  Backward $\overleftarrow{\mathbf{h}}_n \leftarrow \text{LSTM}_{bwd}[\bar{\mathbf{f}}_n, \overleftarrow{\mathbf{h}}_{n+1}]$
  Concatenate: $\mathbf{h}_n = [\overrightarrow{\mathbf{h}}_n; \overleftarrow{\mathbf{h}}_n]$
end for
Step 2: Linear Attention Mechanism
Compute energy scores e_n, normalized weights α_n, and context vector $\mathbf{c}_k$.
Step 3: Prediction and Metric
Predict next state: $\hat{\mathbf{y}}_{k+1} = \mathbf{W}_{fc}\mathbf{c}_k + \mathbf{b}_{fc}$
Denormalize: $\widehat{\mathbf{P}}_{lstm,k+1}^{(j)} \leftarrow \text{DeNorm}(\hat{\mathbf{y}}_{k+1})$
Calculate Divergence: $\delta_k^{(j)}$
return $\widehat{\mathbf{P}}_{lstm,k+1}^{(j)}, \delta_k^{(j)}$

B. Q-Learning Based Dwell Time Allocation and Route Planning Algorithm

To address the resource constraints and dynamic geometry problems in multi-object tracking, we propose a closed-loop cognitive architecture, which is shown in Algorithm 2. The system employs a Q-Learning agent to minimize tracking error.

In Q-Learning, $\mathcal{S}_k$ is a composite of the individual states of all targets. To mitigate the curse of dimensionality, we discretize the state $s_k^{(j)}$ of the j-th target into four hierarchical levels: Level 0(Steady,$\sigma_k^{(j)} \leq \sigma_{enter}$, $\delta_k^{(j)} \leq \delta_{enter}$), Level 1(Warning, $\sigma_k^{(j)} \leq \sigma_{enter}$,$\delta_k^{(j)} > \delta_{enter}$), Level 2(Maneuver,$\sigma_k^{(j)} > \sigma_{enter}$), Level 3(Lost,$\sigma_k^{(j)} > \sigma_{panic}$).This classification relies on the tracking error covariance trace $\sigma_k^{(j)}$ from the IMM filter and $\delta_k^{(j)}$ derived in Section III-A.

To prevent rapid state flickering, we apply Schmidt triggers with distinct enter/exit thresholds ($\sigma_{enter} = 55.0m, \sigma_{exit} = 40.0m, \delta_{enter} = 22.0m, \delta_{exit} = 14.0m, \sigma_{panic} = 150.0m$).

To handle continuous time, we discretize the action space using Simplex Sampling. The total time budget is divided into $N_{res} = 6$ units. An action a_k assigns $n_k^{(j)}$ units to target j, satisfying $\sum_{j=1}^{M} n_k^{(j)} = N_{res}$ with minimum guarantee $n_k^{(j)} \geq 1$. These integer allocations are normalized to priority weights $w_k^{(j)} = n_k^{(j)}/N_{res}$, which then scale the physical beam dwell time to $t_k^{(i,j)} = \Delta t \cdot w_k^{(j)}$.

We design a hybrid reward function that combines a linear term for large errors and a logarithmic term for fine-grained precision. The global reward is the aggregation of individual penalties $R_k = \sum_{j=1}^{M} r_k^{(j)}$. The immediate reward $r_k^{(j)}$ is defined as:

$$r_k^{(j)} = \begin{cases} -1000, & \text{if } \sigma_k^{(j)} > \sigma_{panic} \\ -\omega_k^{(j)} \Big[0.5\xi_k^{(j)} \\ \qquad + 5\log_{10}(1+\xi_k^{(j)}) \Big], & \text{otherwise} \end{cases} \tag{6}$$

where $\xi_k^{(j)} = \|\widehat{\mathbf{P}}_{t,k}^{(j)} - \mathbf{P}_{t,k}^{(j)}\|_2$ represents the IMM estimate error for the j-th target. $\omega_k^{(j)} \in \{0.3, 2.0, 5.0, 10.0\}$ serves as a context-aware scaling factor matched to states (from Level 0 to Level 3) to avoid reward hacking.

Coefficients in (6) were empirically determined. The Q-learning agent's autonomous optimization within the MDP framework ensures policy robustness and minimizes the impact of initial human-selected values.

The agent maintains a Q-table $Q(\mathcal{S}, a)$ and updates it iteratively using the Bellman equation. An ϵ-greedy strategy is employed to select a_k for exploration.

For the i-th radar, the system identifies the primary target index,denoted as j^*,which consumes the maximum allocated dwell time:

$$j^* = \arg \max_{j \in \{1,\ldots,M\}} \left(t_k^{(i,j)} \right) \tag{7}$$

Once j^* is identified, the radar employs a Line-of-Sight (LOS) Guidance strategy to maximize the SNR. The guidance vector $\mathbf{u}_k$ and the relative distance U_k are calculated as:

$$\mathbf{u}_k = \widehat{\mathbf{P}}_{t,k}^{(j^*)} - \mathbf{P}_{r,k}^{(i)}, \quad U_k = \|\mathbf{u}_k\|_2 \tag{8}$$

To prevent collisions and numerical singularities, we strictly enforce a safety zone. The velocity $\mathbf{v}_{r,k}^{(i)}$ is computed to drive the radar towards the target at maximum cruising speed $V_{max} = 50.0$ m/s, subject to a safety distance threshold $R_{safe} = 2000.0$ m:

$$\mathbf{v}_{r,k}^{(i)} = \begin{cases} V_{max} \cdot \frac{\mathbf{u}_k}{U_k}, & \text{if } U_k > R_{safe} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

Algorithm 2 Q-Learning

Require: Global State $\mathcal{S}_k$, Q-table $Q(S, A)$, Previous Allocation ζ_{k-1}

Ensure: Dwell Time $\zeta_k = \{t_k^{(i,j)}\}$, Radar Velocities $\mathbf{v}_{r,k}^{(i)}$

Step 1: State Perception
for each target j **do**
    Calculate metrics $\sigma_k^{(j)}, \delta_k^{(j)}$ and update state $s_k^{(j)}$
end for
Construct Global State $\mathcal{S}_k$
Step 2 & 3: Action Decision & Allocation
Select $a_k \leftarrow \epsilon-\text{greedy}(Q(\mathcal{S}_k, \cdot))$
Compute global priority weights $w_k^{(j)}$
for each radar i and target j **do**
    Calculate dwell time $t_k^{(i,j)}$
end for
Step 4: Route Planning
for each radar $i = 1$ to N **do**
    Identify primary target: j^*
    if $t_k^{(i,j^*)} > 0$ **then**
        Set velocity $\mathbf{v}_{r,k}^{(i)}$ towards target j^*
    else
        $\mathbf{v}_{r,k}^{(i)} \leftarrow 0$
    end if
end for
return $\zeta_k, \mathbf{v}_{r,k}^{(i)}$

C. Cognitive Tracking Cycle

At each time step k, the system fuses distributed measurements to simultaneously update the IMM estimator and the Bi-LSTM maneuver detector. These outputs are aggregated into the global state $\mathcal{S}_k$, which the Q-Learning agent maps to a joint action. By coupling electronic dwell time allocation with physical radar pursuit, the system actively optimizes the observation geometry, thereby minimizing the uncertainty for the subsequent step $k+1$.

IV. Simulation Results

To validate the effectiveness of the proposed cognitive trac king architecture, we conducted Monte Carlo simulations ($N_{mc} = 200$).

A. Simulation Setup

We consider $N = 3$ radars initially positioned at $(2, -3), (-4, 12)$and$(10, 10)km$, tracking $M = 3$ targets over $150s$. Target 1 performs constant velocity (CV) motion, while

Targets 2 and 3 execute Coordinate Turn (CT) maneuvers to simulate single-turn and complex S-shape trajectories, respectively. The targets are initialized with a velocity of $(v_x, v_y) = (80, 50)m/s$.The IMM filter's model set $\mathbb{M}$ includes CV and CT models with turn rates of $\pm 2.0°$/s.Radar visibility is capped at $25km$.

The Transition Probability Matrix Π of the IMM filter governs the mode switching:

$$\Pi = \begin{bmatrix} 0.90 & 0.05 & 0.05 \\ 0.05 & 0.90 & 0.05 \\ 0.05 & 0.05 & 0.90 \end{bmatrix} \tag{10}$$

B. Tracking Accuracy Analysis

We compare the tracking RMSE of the proposed Cognitive Resource Allocation (CRA) against the Uniform Allocation Baseline (UAB). The results are presented in Fig. 1, Fig. 2, and Fig. 3.

For steady Target 1, CRA maintains $RMSE \approx 20m$ compared to $100m$ in UAB. During maneuvers, UAB suffers from model lag with errors peaking at $150m$ (Target 2) and $200m$ (Target 3). Conversely, the proposed architecture keeps peak errors below $50m$ and $80m$, respectively, by actively reducing the measurement noise variance through physical pursuit.

C. Dwell Time Allocation and Route Planning

To evaluate the decision-making logic of the Q-Learning agent, we visualize the temporal distribution of beam resources in Fig. 4. Note that the heatmap visualizes the Cumulative Dwell Time, defined as the sum of dwell times across all N radars, so the value represents the total system energy focus.

The color intensity represents the dwell time allocated to each target at each time step, with yellow indicating high priority and dark blue indicating low priority. The heatmap reveals a distinct contrast: minimal resources are wasted on Target 1, whereas allocation saturates for Target 2 and Target 3.

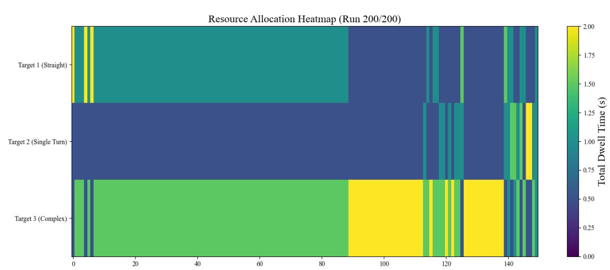

Fig. 4. Resource Allocation Heatmap.

Fig. 5 shows radars curving to intercept high-threat regions, enhancing SNR via range reduction.This spatial reconfiguration directly explains the superior tracking accuracy and robustness observed in the preceding RMSE analysis.

Table I justifies the necessity of the IMM filter. While pure Bi-LSTM offers precise point estimation, it lacks the error covariance required to quantify observation uncertainty. Crucially, the IMM's covariance trace is indispensable for defining

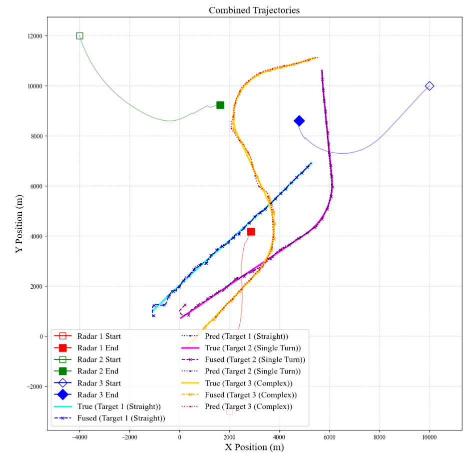

Fig. 5. Combined Trajectories.

the Q-learning state space $\mathcal{S}_k$. Our hybrid architecture balances these strengths: using the IMM as a statistical foundation for resource allocation and Bi-LSTM to compensate for model lag, thereby minimizing peak tracking errors during maneuvers.

TABLE I
ABLATION STUDY: TRACKING RMSE COMPARISON (M)

Target	Metric	Pure IMM	Pure Bi-LSTM	Proposed Hybrid
1	Peak RMSE	395.0	390.0	390.5
	Mean RMSE	95.3	25.1	35.4
2	Peak RMSE	420.0	370.0	372.0
	Mean RMSE	90.5	25.8	35.6
3	Peak RMSE	320.4	310.0	310.2
	Mean RMSE	130.2	38.5	45.1

V. CONCLUSION

We proposed a cognitive cooperative tracking architecture integrating Bi-LSTM active perception with Q-Learning. The Bi-LSTM module effectively anticipates maneuvers, while the Q-Learning agent jointly optimizes dwell time and platform motion to enhance observation geometry. Simulations confirm that this closed-loop mechanism significantly outperforms static baselines, reducing peak errors through active physical pursuit. Future work will extend this framework to 3D scenarios with continuous control and anti-jamming capabilities.

This work was supported in part by the Basic Science (Natural Science) Research Project of Higher Education Institutions in Jiangsu Province under grant 25KJB510015, in part by the Startup Foundation for Introducing Talent of Nanjing University of Information Science and Technology, China (no. 1513142601022), and in part by Key Laboratory of Radar Imaging and Microwave Photonics (Nanjing University of Aeronautics and Astronautics), Ministry of Education, No. NJ20250003.

REFERENCES

[1] S. Haykin, "Cognitive radar: a way of the future," in *IEEE Signal Processing Magazine*, vol. 23, no. 1, pp. 30-40, Jan. 2006.

[2] G. E. Smith, S. Z. Gurbuz, S. Brüggenwirth and P. John-Baptiste, "Neural Networks & Machine Learning in Cognitive Radar," in *2020 IEEE Radar Conference (RadarConf20)*, Florence, Italy, 2020.

[3] Z. Lu and M. C. Gursoy, "Resource Allocation for Multi-Target Radar Tracking via Constrained Deep Reinforcement Learning," in *IEEE Transactions on Cognitive Communications and Networking*, vol. 9, no. 6, pp. 1677-1690, Dec. 2023.

[4] Z. Lu, M. C. Gursoy, C. K. Mohan and P. K. Varshney, "Learning-Based Cognitive Radar Resource Management for Scanning and Multi-Target Tracking," in *ICC 2024 - IEEE International Conference on Communications*, Denver, CO, USA, 2024, pp. 1-6.

[5] Z. Lu, M. C. Gursoy, C. K. Mohan and P. K. Varshney, "Adaptive Resource Management in Cognitive Radar via Deep Deterministic Policy Gradient," *2025 IEEE International Radar Conference (RADAR)*, Atlanta, GA, USA, 2025, pp. 1-6.

[6] Q. Wang, Y. Qiao and L. Gao, "A Cognitive Radar Waveform Optimization Approach Based on Deep Reinforcement Learning," in *IEEE International Conference on Signal Processing*, 2018.

[7] J. Zhu et al., "Adaptive Selection of Cognitive Radar Waveforms Based on Improved Deep Q-Learning Network," *2023 3rd International Conference on Computer Science (CEI)*, Wuhan, China, 2023, pp. 515-522.

[8] Z. Lu, S. Kalia, M. C. Gursoy, C. K. Mohan and P. K. Varshney, "Multi-Objective Reinforcement Learning for Cognitive Radar Resource Management," *ICASSP 2025*, Hyderabad, India, 2025, pp. 1-5.

[9] A. Lei, W. Fan and F. Zhou, "A Cognitive Radar Anti-Jamming Strategy Generation Algorithm based on Dueling Double DQN," *2023 IEEE International Radar Conference (RADAR)*, Sydney, Australia, 2023, pp. 1-5.

[10] C. E. Thornton, M. A. Kozy, R. M. Buehrer, A. F. Martone and K. D. Sherbondy, "Deep Reinforcement Learning Control for Radar Detection and Tracking in Congested Spectral Environments," in *IEEE Transactions on Cognitive Communications and Networking*, vol. 6, no. 4, pp. 1335-1349, Dec. 2020.

[11] C. Gao et al., "Maneuvering Target Tracking with Recurrent Neural Networks for Radar Application," *2018 International Conference on Radar (RADAR)*, Brisbane, QLD, Australia, 2018, pp. 1-5.

[12] C.-Y. Wei, "Track-LITE: Low-Integration Threshold Enhancements for Energy-Efficient Target Tracking With Dynamic Triggering," in *IEEE Transactions on Vehicular Technology*, 2025.

[13] K. Shen, W. Yuan, J. Yan and K. Ma, "MCST: An Adaptive Tracking Algorithm for High-Speed and Highly Maneuverable Targets Based on Bidirectional LSTM Network," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 2, pp. 3205-3226, April 2025.

[14] P. Valianti, K. Malialis, P. Kolios and G. Ellinas, "Cooperative Search and Track of Rogue Drones using Multiagent Reinforcement Learning," *2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Kuching, Malaysia, 2024, pp. 2091-2096.

[15] J. Cheng, K. Wang, H. Shin, T. Q. S. Quek and M. Win, "Reinforcement Learning for Joint Radar and Communication Enabled Multi-UAV Cooperative Dynamic Target Detection and Tracking," in *IEEE Transactions on Vehicular Technology*, early access, doi: 10.1109/TVT.2025.3597638.